DriveCompose: Generative Reconstruction of Compositional Driving Scenes

Anonymous CVPR submission

Paper ID *****

Abstract

Reusing a driving scene under new camera and actor states exposes surfaces that were never recorded. Reconstructing such a scene requires recovering the observed content, completing missing geometry and appearance, and retaining a stable identity for each movable component. We present DriveCompose, a framework for generative reconstruction of compositional driving scenes from calibrated image sequences. A shared visual encoder estimates geometry, actor identities, and motion. We aggregate this evidence into world-space background queries and canonical vehicle and human queries, then encode them in a structured latent representation. A conditional flow model completes unknown codes while exchanging context across scene components. Observed codes and known free space constrain this generation process. Type-specific decoders produce Gaussian background and assets, which are checked against the input views before use. Learning proceeds from visual recovery and latent representation fitting to completion-prior training and adaptation with rendered observations. The reconstructed scene is subsequently composed under explicit camera, rigid-body, and skeletal states. This formulation gives generation a persistent three-dimensional output: the same completed content can be rendered along different trajectories without resampling its geometry or appearance.

1. Introduction

Driving simulation and scene editing require a reconstruction that remains usable when the recorded configuration changes. Moving a camera reveals a road patch hidden behind a vehicle. Turning that vehicle exposes its unseen side, and changing a pedestrian's pose brings previously occluded body surfaces into view. These operations require geometry and appearance beyond the recorded images, together with a representation that keeps the content attached to the same scene and actor identities.

Recent feed-forward methods recover driving scenes directly from temporal observations. STORM aggregates dynamic Gaussians through estimated scene flow, and DGGT jointly recovers camera parameters and dynamic scene content from unposed images [1, 14]. StreetForward and PointForward further study temporal processing and the organization of reconstructed evidence [3, 16]. For an asset that will later be moved or articulated, however, observations across time must also be gathered into a stable local representation, and the surfaces exposed by the new state must be supplied.

Compositional representations address the first requirement. Street Gaussians separates vehicles from the background; OmniRe includes canonical models for rigid and articulated actors; HUGSIM connects scene content to control-driven state updates [2, 12, 17]. Generative models address missing content through asset priors, Gaussian completion, or generated views [6, 13, 15, 18]. Combining them requires a generator that assigns content to the background and to individual assets. A generated surface must have an owner, a coordinate frame, and a motion rule before it can become part of an editable scene.

We formulate driving-scene reconstruction as conditional generation of a compositional scene. Our representation follows the units that will be manipulated: a world-space background, rigid vehicles in local coordinates, and humans on canonical surfaces with skeleton bindings. Image evidence is first aggregated on these supports. A generative model then completes their missing latent content under the observed scene conditions. Each completed surface is decoded in the coordinate frame used to control its component.

DriveCompose implements this formulation with a visual recovery front end, structured latent codes, and a conditional flow model (Fig. 1). The front end estimates geometry, trajectory-level identities, and time-varying states. Canonical aggregation combines observations of the same content across views and time. Within the completion model, local attention follows each component's support, while shared memory exchanges context between background and assets. Each query carries both its canonical coordinate and its placement in a reference world configuration: the former identifies persistent content, and the latter expresses its spatial relation to the scene. Type-specific decoders convert the completed codes into Gaussian components.

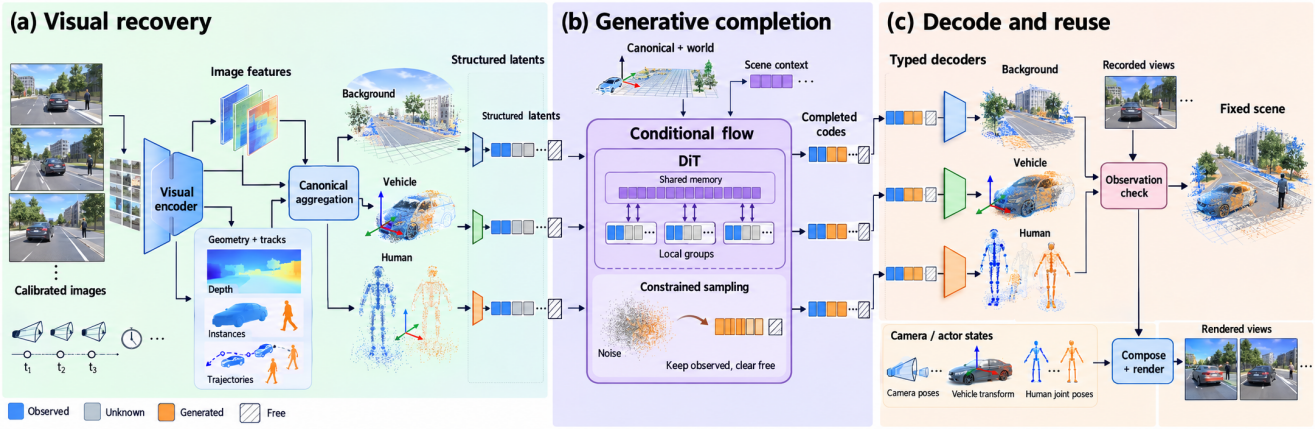


Figure 1. **DriveCompose overview.** (a) Calibrated images provide geometry, identity, and motion evidence, which is aggregated on background and canonical asset supports and encoded into structured latents. (b) A scene-conditioned flow transformer completes unknown codes while retaining observed content and suppressing free space. (c) Typed decoders produce Gaussian components. A check in the original views precedes scene reuse; camera and actor states then control composition and rendering. Scene depictions are schematic, not experimental outputs.

Generation must also respect the images from which the scene was recovered. We retain reliable observed codes and suppress known free space throughout sampling. Newly generated content can nevertheless occlude an observed surface, so we check the assembled candidate in the original views as well. After construction, the background and canonical assets remain fixed. Camera placement, vehicle transforms, and human joint poses determine their subsequent rendering. Learning follows this separation: we fit recovery and the latent representation before training the completion prior, then adapt generation through differentiable rendering while keeping the representation fixed.

Our contributions are:

- A generative reconstruction framework that organizes driving observations into a compositional Gaussian scene, with world-space background and persistent canonical assets for state-driven reuse.
- A structured completion model that combines category-specific supports and scene-wide context with observed-code constraints, free-space suppression, and rendered observation checks.

2. Related work

Learned driving-scene reconstruction. Feed-forward models replace repeated scene fitting with learned inference from image observations. STORM predicts Gaussians and scene flow for dynamic outdoor reconstruction [14]. DGGT adds pose-free reconstruction, temporal visibility, and motion-guided interpolation [1]. StreetForward, PointForward, and Instant NuRec explore temporal processing, point-aligned representations, and feed-forward reconstruction for simulation [3, 4, 16]. We build on the role of image-based recovery as a source of geometry and motion evidence. Our output variables are background and canonical asset codes, which support conditional completion before the scene is rendered under new states.

Compositional scenes and control. Street Gaussians models tracked vehicles separately from static background [12]. OmniRe extends this organization to multiple actor types with category-appropriate motion models [2]. HUGSIM provides control-driven state updates and interactive simulation using reconstructed content [17]. These methods separate persistent appearance and geometry from motion. We retain this separation within the generative representation, so completed surfaces inherit an asset identity and its rigid or articulated transform.

Generative reconstruction and completion. StreetCrafter uses three-dimensional conditions to generate views that guide reconstruction, and HorizonForge combines editable scene content with video diffusion [10, 13]. WorldSplat generates pixel-aligned Gaussians from a latent model and refines their rendered videos [18]. GenAssets learns priors for driving assets, CompleteSplat completes Gaussian scenes from image evidence, and JRM uses latent flow matching to combine unaligned observations of shared objects [6, 11, 15]. LatentWorld organizes semantic occupancy generation around scene entities [8]. DriveCompose conditions on a recovered driving log and completes RGB-renderable background and canonical assets in one query layout. Generation acts on their unknown latent content; observed evidence constrains both sampling and the assembled render.

3. Method

DriveCompose reconstructs a driving scene in the units that will later be controlled. We first recover visual evidence and aggregate it on background and canonical asset supports (Secs. 3.1–3.2). A typed codec defines the latent variables and their Gaussian decoding (Sec. 3.3). Conditional flow completes missing content in this representation (Sec. 3.4), after which the scene is composed and checked in the observed views (Sec. 3.5). Section 3.6 describes how these modules are learned. Figure 1 shows the complete construction and reuse path.

3.1. Problem formulation and compositional representation

The input is a synchronized log of images $\mathbf{I}_{t,v}$, timestamps, intrinsics $\mathbf{K}_v$, and camera-to-rig calibration $\mathbf{T}^r_{c_v}$. Rig poses $\mathbf{T}^w_r(t)$ are provided or estimated relative to the first frame, giving camera poses $\mathbf{T}^w_{c_v}(t) = \mathbf{T}^w_r(t)\mathbf{T}^r_{c_v}$. Transforms act on column vectors and distances are in meters. Images follow a rectified pinhole model and linear RGB; distortion, rolling shutter, and exposure differences are handled before evidence aggregation.

The output is a compositional Gaussian scene

$$\mathcal{W} = \big(\mathcal{G}_b, C_{\text{sky}}, V_{\text{sky}}, \{\mathcal{A}_a\}, \mathcal{B}, \xi\big), \tag{1}$$

where $\mathcal{G}_b$ is the world-space background, C_{sky} a sky cubemap, V_{sky} its supported directions, $\mathcal{A}_a$ the canonical asset for identity a, $\mathcal{B}$ the finite spatial support, and ξ the sampling seed. A Gaussian stores its center $\boldsymbol{\mu}$, covariance $\boldsymbol{\Sigma}$, opacity α, and directional appearance, and is rendered by Gaussian splatting [5]. Vehicles retain local geometry and dimensions. Humans additionally retain a rest skeleton and skinning weights. Scene state controls the placement and articulation of these fixed components.

We model missing content through conditional generation. Let $\mathcal{Q}$ be the recovered query layout and C its observed evidence. The completion model samples unknown codes $\mathbf{z}_u$, while the codec D_ψ decodes the assembled codes into scene content:

$$\mathbf{z}_u \sim p_\theta(\mathbf{z}_u \mid C, \mathcal{Q}), \qquad \mathcal{W} = D_\psi(\mathbf{z}^{\text{obs}}, \mathbf{z}_u; \mathcal{Q}). \tag{2}$$

The decoder also carries forward the fixed support, identity, binding, and sky-validity metadata in Eq. 1. The layout is established from the input log; generation fills its unknown content. Simulation time changes the state of the decoded scene, while flow time indexes the construction of its latent codes.

3.2. Visual recovery and canonical aggregation

Shared visual encoding. The Driving Log Encoder converts images to patch tokens and adds image positions, camera calibration, and time embeddings. Within-image attention extracts spatial features. A second transformer processes pooled image summaries across views and timestamps, and returns the resulting context to the patch features. Geometry heads predict camera-axis depth, normals, confidence, semantics, and instance assignments. The confidence determines which predictions can contribute surface evidence.

Learned actor queries attend to all image tokens to form trajectory-level identity slots. Each slot then reads the features at an individual timestamp to estimate its state. This separates persistent identity, dimensions, and human shape from frame-dependent root poses and joint rotations. Root poses are predicted in the rig frame and mapped to world coordinates with $\mathbf{T}^w_r(t)$. A human shape code selects an identity-specific template with fixed canonical support, skeleton topology, and binding. Slot capacity bounds the number of represented actors.

Query supports. A query stores canonical position $\mathbf{q}_i$, cell scale $\boldsymbol{\delta}_i$, type k_i, local attention group g_i, and reference-world coordinate $\mathbf{q}^w_i$. Reliable depth samples provide surface anchors. Background anchors are combined with a finite world grid; vehicle anchors are motion-normalized and combined with a grid in the local bounding volume. Human queries lie on the canonical template surface. Sky queries are unit directions grouped by cubemap face. These supports provide explicit locations for both observed and missing content.

The layout remains fixed during completion. We obtain $\mathbf{q}^w_i$ by placing the query in the first input configuration, including skinning for humans. Canonical coordinates identify the content within its component; reference-world coordinates let the generator relate that component to its surroundings.

Motion-aware evidence aggregation. For each observation, we place the canonical query at the corresponding actor state:

$$\mathbf{p}_i(t) = \begin{cases} \mathbf{q}_i, & k_i = \text{background}, \\ \text{xyz}(\mathbf{T}^w_a(t)\bar{\mathbf{q}}_i), & k_i = \text{vehicle}, \\ \text{xyz}(\mathbf{T}^w_h(t)\mathbf{A}_i(t)\bar{\mathbf{q}}_i), & k_i = \text{human}. \end{cases} \tag{3}$$

Here $\bar{\mathbf{q}}_i$ is homogeneous, xyz extracts Euclidean coordinates, and $\mathbf{A}_i(t)$ is the weighted skinning transform. Projection into a camera gives depth $z_{i,t,v}$ and sampled feature, depth, and confidence $(\boldsymbol{\phi}_{i,t,v}, d_{i,t,v}, c_{i,t,v})$. With tolerance $\eta = \eta_{\text{abs}} + \eta_{\text{rel}} d$, define $r = (d - z)/\eta$. A surface observation must lie inside the image, have valid depth and sufficient confidence, match the query's instance, and satisfy $|r| \le 1$.

Background queries additionally require static-background semantics. Let $a_{i,t,v}$ denote this validity test. We aggregate

$$w_{i,t,v} = a_{i,t,v} c_{i,t,v} \exp(-r_{i,t,v}^2/2), \tag{4}$$

$$\mathbf{h}_i = \frac{\sum_{t,v} w_{i,t,v} [\boldsymbol{\phi}_{i,t,v}; r_{i,t,v}]}{\max(\varepsilon, \sum_{t,v} w_{i,t,v})}. \tag{5}$$

The query confidence is the maximum over accepted observations. The test excludes observations whose depth or instance identity disagrees with the query. A reliable ray votes for free space only when the entire support cell is in front of the measured surface:

$$z_{i,t,v} + \tfrac{1}{2}\|\boldsymbol{\delta}_i\|_2 < d_{i,t,v} - \eta. \tag{6}$$

Positive accumulated observation weight marks a query as observed. Otherwise, sufficient free-space votes mark it free, and the remainder is unknown. The masks m_i, f_i, and u_i are mutually exclusive and satisfy $m_i + f_i + u_i = 1$. Sky projection uses camera rotation alone, reliable sky semantics, and zero depth residual; sky queries receive no free-space votes.

3.3. Structured latent representation and decoding

Type-specific encoders map $[\mathbf{h}_i; \mathbf{q}_i; \boldsymbol{\delta}_i]$ to codes $\mathbf{z}_i \in \mathbb{R}^D$. Each query keeps its type, ownership, and coordinate frame even though all codes have the same dimension. Visibility masks and confidence are supplied separately to the completion model. Partial and complete observations therefore use the same intrinsic encoding map. Before conditioning, unobserved codes are set to zero; the observed subset is denoted $\mathbf{z}^{\text{obs}}$.

Background, vehicle, and human decoders independently map each query to K child Gaussians. For child k of query i, the center is confined to its support cell:

$$\boldsymbol{\mu}_{i,k} = \mathbf{q}_i + \tfrac{1}{2}\boldsymbol{\delta}_i \odot \tanh(\mathbf{b}_{i,k}). \tag{7}$$

Positive scales and normalized rotations determine positive-definite covariance. Opacity is multiplied by $1 - f_i$, so free queries cannot produce visible Gaussians. Human children inherit the skinning weights of their support query; vehicle children share their asset's rigid transform. A separate sky head predicts cubemap colors, with the observation-supported domain retained in V_{sky}.

Appearance uses a constant RGB term and three local directional coefficients per channel, evaluated as $c(d) = \max(0, c_0 + C_{\text{app}} \hat{d}_{\text{local}})$. Decoding is per query: once the codec is frozen, an unchanged observed code produces unchanged Gaussian parameters. The completion network can exchange context across queries without changing this decoding property.

3.4. Conditional generative reconstruction

Scene-conditioned transformer. We parameterize the completion velocity with a DiT-style transformer [9] trained by flow matching [7]. Its input combines current and observed codes, masks, confidence, type, both coordinate descriptions, cell scales, and pooled log features. Background groups follow spatial neighborhoods, each asset forms a local group, and sky directions are grouped by cubemap face. Local self-attention processes these groups separately. Shared memory then communicates their context:

$$\mathbf{M}' = \mathbf{M} + \text{Attn}_r(\mathbf{M}, \widetilde{\mathbf{X}}, \widetilde{\mathbf{X}}), \tag{8}$$

$$\mathbf{U} = \widetilde{\mathbf{X}} + \text{Attn}_w(\text{LN}(\widetilde{\mathbf{X}}), \mathbf{M}', \mathbf{M}'). \tag{9}$$

Here $\widetilde{\mathbf{X}}$ concatenates locally updated tokens, and attention arguments are queries, keys, and values. A residual feed-forward update follows $\mathbf{U}$. Flow-time modulation enters the local normalization; learned tokens and log summaries initialize memory. This structure keeps each support's local organization while allowing a vehicle or human query to use information from the surrounding scene.

Flow training and constrained sampling. Let $\mathbf{z}_1$ be a complete target on the same query layout and $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$ be Gaussian noise. We interpolate only the unknown content at flow time $\tau \in [0, 1]$:

$$\mathbf{z}_\tau = m \odot \mathbf{z}^{\text{obs}} + u \odot [(1 - \tau)\boldsymbol{\epsilon} + \tau \mathbf{z}_1]. \tag{10}$$

Observed codes remain fixed and free-space codes remain zero. The target velocity for unknown queries is $\mathbf{z}_1 - \boldsymbol{\epsilon}$. With an optional loss selector s_i, the objective is

$$\mathcal{L}_{\text{flow}} = \frac{\sum_i u_i s_i \|\mathbf{v}_\theta(\mathbf{z}_\tau, \tau, C)_i - (\mathbf{z}_{1,i} - \boldsymbol{\epsilon}_i)\|_2^2}{D \max(1, \sum_i u_i s_i)}. \tag{11}$$

All unknown queries interpolated in Eq. 10 require valid target codes, even if s_i excludes some of them from the loss. This requirement determines how complete targets are constructed in Sec. 3.6.

At inference, we initialize unknown codes with noise and take S Euler steps with $\Delta\tau = 1/S$:

$$\mathbf{z}^{j+1} = \Pi_{m,f}\left[\mathbf{z}^j + \Delta\tau\, u \odot \mathbf{v}_\theta(\mathbf{z}^j, j/S, C)\right]. \tag{12}$$

The projection $\Pi_{m,f}$ restores observed codes and clears free-space codes after every step. The seed fixes one sampled completion. Sampling precedes decoding, and the resulting scene is reused when the camera or actor states change.

3.5. Scene composition and observation consistency

State-driven rendering. At online step k, state s_k specifies the ego pose, active assets, vehicle transforms, and human root and joint poses. A vehicle's Gaussian center and covariance transform as

$$\boldsymbol{\mu}^w = \mathbf{R}\boldsymbol{\mu}^a + \mathbf{t}, \qquad \boldsymbol{\Sigma}^w = \mathbf{R}\boldsymbol{\Sigma}^a\mathbf{R}^\top. \tag{13}$$

For humans, the weighted skinning transform is $\mathbf{A}_i = \sum_j w_{ij}\mathbf{P}_j\mathbf{B}_j^{-1}$, where $\mathbf{B}_j$ and $\mathbf{P}_j$ are rest and current global joints in the root frame. The transformed center is $\bar{\boldsymbol{\mu}}_i^w = \mathbf{T}_h^w\mathbf{A}_i\bar{\boldsymbol{\mu}}_i^h$. The combined local linear map $\mathbf{J}_i$ gives covariance $\mathbf{J}_i\boldsymbol{\Sigma}_i\mathbf{J}_i^\top$. Its inverse $\mathbf{J}_i^{-1}$ maps viewing directions into the canonical frame; the direction is normalized before appearance evaluation. Nearly singular skinning transforms are rejected. The root pose controls overall placement, while the root joint retains its rest transform.

The renderer composites the background, active assets, and valid sky. External policy and dynamics can advance the state:

$$o_k = \mathcal{R}(\mathrm{Compose}(\mathcal{W}, s_k), \mathcal{C}_k), \tag{14}$$

$$s_{k+1} = F_{\mathrm{dyn}}(s_k, \pi(o_k), \Delta t). \tag{15}$$

Resetting or branching a trajectory changes state while preserving $\mathcal{W}$ and its seed. Finite support and supported sky directions determine coverage. Road meshes and collision proxies share the visual assets' coordinates and are maintained by the physical backend.

Checking a constructed scene. Fixed observed codes preserve decoded parameters, but a newly generated surface can still occlude them. Before accepting a scene, we render the completed candidate and an observed-only reference at the original cameras and actor states. The reference disables all non-observed queries. Let $e_{\mathrm{cand}}(p)$ and $e_{\mathrm{obs}}(p)$ be their mean absolute RGB errors at pixel p. We measure local degradation with

$$\Delta e(p) = \max\{e_{\mathrm{cand}}(p) - e_{\mathrm{obs}}(p), 0\}. \tag{16}$$

Averaging this quantity over reliable non-ego pixels prevents improvements in one region from cancelling degradation elsewhere. Acceptance also bounds absolute RGB and valid-depth errors, minimum coverage, and foreground intrusion, using aggregate and worst-view checks. An intrusion is a newly rendered surface in front of the predicted reference depth that is absent from the observed-only render; its rate is normalized over reliable non-sky geometry pixels. Predicted depth provides an observation-consistency reference. Independent geometry evaluation requires a separate reference. A rejected candidate is not exported for reuse.

3.6. Training procedure

Learning is divided into four stages so that the completion target has a stable meaning. Recovery determines the evidence and support, the codec defines its latent coordinates, and the completion network learns a distribution over the missing codes. The last stage adapts this distribution to real observations without changing the representation it predicts.

Stage 1: visual recovery. We supervise the image encoder, geometry heads, identity slots, and state heads using geometry, semantics, instance assignments, and actor annotations. One trajectory-level matching assigns slots to target identities and is reused at every timestamp. Root translation, rotation, and fixed dimensions constrain vehicle states; humans additionally require identity-consistent shape, template binding, and joint supervision. Confidence is trained or calibrated against geometric reliability, using depth, occlusion, and reprojection information. These targets establish the coordinate and identity consistency required by aggregation.

Stage 2: latent representation fitting. We freeze recovery and train the type-specific encoders and decoders. For each sample, partial input establishes the query layout once. More complete observations are then aggregated on exactly the same positions, types, order, and ownership. This correspondence is required for latent completion: separately rebuilding a complete-scene layout would change the variables being predicted. RGB, depth, and silhouette supervision train the decoded scene, with empty-space opacity and Gaussian-scale regularization. Gaussian-set targets can also be matched as sets; direct coordinate supervision is used only when child correspondences are available.

For rendered image $\widehat{\mathbf{I}}$ and depth $\widehat{d}$, the observation terms used in representation fitting and later adaptation are masked L1 errors:

$$\mathcal{L}_{\mathrm{rgb}} = \langle \|\widehat{\mathbf{I}} - \mathbf{I}\|_1/3 \rangle_{\Omega_I}, \tag{17}$$

$$\mathcal{L}_{\mathrm{depth}} = \langle |\widehat{d} - d| \rangle_{\Omega_d}. \tag{18}$$

The notation $\langle \ell \rangle_\Omega$ denotes a mean over valid elements of mask Ω. The depth target d and its validity mask are supplied by the available supervision; they do not make the observation check an independent accuracy measurement.

Stage 3: completion-prior learning. We freeze recovery and the entire codec, encode complete targets as $\mathbf{z}_1$, and optimize the completion transformer with Eq. 11. Complete scene or asset examples must provide a valid target for every unknown query in Eq. 10; partial real logs alone do not meet this condition. Complete scenes provide cross-component

context, while complete assets supply within-object content. A validity test rejects targets with unsupported unknown queries.

Conditioning masks simulate contiguous unseen surfaces, background holes, long occlusions, short tracks, and uneven camera coverage. Perturbations to recovered poses, instance assignments, and confidence expose the generator to imperfect evidence. Target codes and input codes retain the same codec and query definition throughout this stage.

Stage 4: adaptation through rendering. Keeping recovery and the codec frozen, we differentiate through flow integration, decoding, and rendering to update only the completion network. Partial real logs supervise the rendered observations. Known-free rays and geometric backend samples constrain added content:

$$\mathcal{L}_{\text{scene}} = \lambda_e \langle a_r^2 \rangle + \lambda_g \langle [-d_f]_+^2 \rangle + \lambda_c \langle |d_f| \rangle_{\Omega_c} + \lambda_p \langle [d_p]_+^2 \rangle, \tag{19}$$

$$\mathcal{L}_{\text{adapt}} = \lambda_r \mathcal{L}_{\text{rgb}} + \lambda_d \mathcal{L}_{\text{depth}} + \mathcal{L}_{\text{scene}}. \tag{20}$$

Here a_r is accumulated opacity along a known-free ray, d_f is foot signed distance positive above ground, Ω_c selects contacting feet, and d_p is penetration depth positive inside another actor. We use $[x]_+ = \max(x, 0)$; unmasked means run over supplied samples. The coefficients λ weight the active constraints. Free-ray opacity constrains the rendered effects of added content beyond the query-level free-space mask.

Complete-target samples can be interleaved with real-log adaptation to retain the learned prior, yielding $\mathcal{L}_4 = \lambda_f \mathcal{L}_{\text{flow}} + \mathcal{L}_{\text{adapt}}$, with each term applied only to samples carrying its required targets. Observed anchors are enforced by frozen decoding and hard code restoration.

Inference. A log is encoded once, its support and evidence are fixed, and the completion model integrates the unknown codes. We then decode and check the scene before exporting its fixed content. Runtime rendering uses only this content and the requested states. Recovery, codec, completion, and query-definition versions are recorded together, since changing any of the first two changes the latent distribution on which completion was learned.

4. Experiments

We study dynamic scene reconstruction from sparse temporal observations along two axes: recovering scene appearance at unobserved times and rendering the reconstructed scene from off-trajectory viewpoints. Temporal evaluation separates interpolation from short-horizon prediction, while spatial evaluation tests rendering quality as the camera moves away from the recorded trajectory.

4.1. Experimental Setup

Dataset and temporal sampling. We evaluate on a fixed subset of 40 sequences from the Waymo Open Dataset scene-flow validation split. We adopt the 20-frame temporal sampling pattern used in STORM and the evaluation resolution used in Instant NuRec [4, 14]. Each clip contains 20 consecutive frames at 10 Hz. We use the FRONT camera and provide four context images at zero-based frame indices $\{0, 5, 10, 15\}$. Temporal interpolation is evaluated at indices $\{1, 2, 3, 4, 6, 7, 8, 9, 11, 12, 13, 14\}$, and temporal extrapolation at $\{16, 17, 18, 19\}$. The latter corresponds to prediction horizons of 0.1–0.4 seconds beyond the last observation. Context frames are excluded from temporal reconstruction scores.

Baselines and inference settings. The current comparison includes STORM, DGGT, and StreetForward using their released checkpoints without additional fine-tuning [1, 14, 16]. All methods observe the same camera and context timestamps and render the same target views. STORM uses calibrated cameras during reconstruction. DGGT and StreetForward estimate cameras and use context-based alignment and intrinsic-calibration adapters before rendering. DGGT is evaluated without diffusion refinement; its temporal adapter extends the last observed tracked motion for extrapolation. These settings define the implementations evaluated here.

Image and depth metrics. RGB predictions and reference images are evaluated at 240×160 pixels. We report PSNR and SSIM over the full image and within shared 2D annotation-box masks. These masks define annotated regions, rather than motion-segmented foreground. For depth, we report AbsRel and δ_1 on valid projected LiDAR samples after per-frame least-squares scale alignment, following the metric formulation in Instant NuRec [4]. Frame scores are averaged within each window, window scores within each sequence, and sequence scores with equal weight across the 40 sequences.

Preprocessing and calibration. The current STORM pipeline resizes quarter-resolution JPEG images to 240×160; DGGT and StreetForward resize the original JPEG images to 518×350 and 504×336, respectively. Each method retains its input normalization. Camera alignment uses the four context observations. In temporal evaluation, one DGGT window uses context-frame LiDAR to resolve an otherwise indeterminate metric scale. In spatial evaluation, DGGT and StreetForward additionally use context-frame LiDAR scale calibration when camera motion is too small or trajectory alignment is unreliable. Calibration excludes the shared annotated regions and uses no target-frame

Table 1. Temporal reconstruction on 40 Waymo validation sequences using one FRONT camera. Each evaluated method covers the same 357 windows. PSNR is in dB; higher is better. † marks one window calibrated with context-frame LiDAR. MOSAIC has not yet been measured.

Method	Interpolation	Extrapolation	All targets
STORM	26.427	25.481	26.191
DGGT†	26.265	25.429	26.056
StreetForward	25.653	24.065	25.256
MOSAIC	–	–	–

depth or RGB.

4.2. Temporal Reconstruction and Prediction

Evaluation protocol. We traverse non-overlapping 20-frame windows throughout each selected sequence. Incomplete or temporally invalid windows are excluded using the shared data checks, yielding 357 valid windows for each method. We report interpolation, extrapolation, and all-target scores separately. The all-target result combines the 12 interpolated frames and four extrapolated frames in each window.

Quantitative comparison. Table 1 reports the measured baseline results. STORM and DGGT achieve all-target PSNR values of 26.191 and 26.056 dB, respectively, while StreetForward achieves 25.256 dB. All three methods obtain lower PSNR on extrapolation than on interpolation. Table 2 complements the full-image comparison with annotated-region scores and scale-aligned depth metrics.

4.3. Off-Trajectory View Synthesis

Evaluation protocol. We evaluate spatial extrapolation on the first 20-frame window of each of the same 40 sequences. Starting from the recorded camera trajectory, we apply fixed lateral offsets of ±1, ±2, and ±4 meters in the local camera frame while retaining camera orientation, intrinsics, and timestamps. Together with the original trajectory, this produces seven 20-frame videos per scene and 280 videos per method. This experiment follows the lateral-shift evaluation strategy of WorldSplat and the extrapolated-view evaluation motivation of PointForward [3, 18].

Distributional evaluation. The shifted trajectories do not have paired reference images. We therefore compute FID and FVD against a fixed population of images and videos from the original recorded trajectories. FID uses the four context timestamps, yielding 160 images per offset and method, to assess spatial changes at observed times. FVD uses the consecutive frames 0–15 at 10 Hz, yielding 40 clips per offset and method and incorporating temporal interpolation. We use Inception image features for FID and I3D video features for FVD. Scores are computed over the pooled sample population for each offset; left and right shifts are reported separately. All methods share the same scenes and real reference population. Lower scores indicate smaller feature-distribution differences; paired visualizations assess scene-specific structure and viewpoint response.

Quantitative comparison. Table 3 reports both metrics. STORM obtains the lowest FID and FVD at the one- and two-meter offsets, whereas StreetForward obtains the lowest values at both four-meter offsets. The ordering therefore depends on the displacement. The original-trajectory scores provide a reference for changes under viewpoint shifts; the spatial results use their own calibrated rendering pipeline and sample scope, rather than the all-window temporal evaluation.

Table 2. All-target image and depth metrics on the same 357 windows. Region scores use shared 2D annotation boxes, not motion-segmentation masks. Depth is evaluated after per-frame scale alignment. The DGGT calibration exception is the same as in Table 1.

	Full image		Annotated regions		Sparse depth	
Method	PSNR ↑	SSIM ↑	PSNR ↑	SSIM ↑	AbsRel ↓	δ_1 ↑
STORM	26.191	0.833	22.083	0.654	0.129	0.867
DGGT†	26.056	0.809	20.809	0.603	0.168	0.779
StreetForward	25.256	0.814	21.844	0.636	0.122	0.860
MOSAIC	–	–	–	–	–	–

Table 3. Off-trajectory synthesis on 40 FRONT-camera clips. FID uses 160 images per trajectory; FVD uses 40 videos of 16 consecutive frames. Each score compares against the same original-trajectory reference population. Left and right denote lateral camera displacement. MOSAIC scores are unmeasured.

FID ↓

Position	STORM	DGGT	SF	Ours
Original	44.56	38.45	50.01	–
Left 1 m	62.50	81.59	92.80	–
Right 1 m	61.44	66.39	75.79	–
Left 2 m	82.92	98.76	111.70	–
Right 2 m	84.52	87.96	94.92	–
Left 4 m	127.56	129.80	125.64	–
Right 4 m	137.58	126.22	121.52	–

FVD ↓

Position	STORM	DGGT	SF	Ours
Original	503.59	707.81	665.42	–
Left 1 m	717.19	939.82	856.76	–
Right 1 m	727.35	1028.18	901.14	–
Left 2 m	1014.69	1189.00	1073.96	–
Right 2 m	1044.03	1205.05	1116.08	–
Left 4 m	1688.40	1710.13	1560.61	–
Right 4 m	1719.92	1613.26	1596.65	–

5. Limitations

DriveCompose represents a finite region under fixed illumination. Errors in recovered depth, identity, or canonical pose can enter the observed anchors and constrain later completion incorrectly. Vehicles are rigid, and humans retain fixed clothing volume and skeleton binding; poses outside the template's support can expose missing geometry. Reference-world coordinates provide context at construction, while contact and collision validity after an edit depend on the physical backend. The observation check assesses agreement with recorded evidence. Geometry outside that evidence and appearance under changed illumination require separate supervision or modeling.

6. Conclusion

We presented DriveCompose, a generative reconstruction framework for compositional driving scenes. Its structured latent representation gathers image evidence into background and canonical assets, giving missing content a defined location, identity, and motion model before generation. Conditional completion uses scene context while retaining observed evidence, and typed decoding produces the components of a reusable Gaussian scene. Separating content construction from state-driven composition allows subsequent camera and actor changes to reuse the same completion.

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